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RESEARCH

Differential Wilting among Soybean Genotypes in Response to Water Deficit

C. Andy King, Larry C. Purcell,* and Kristofor R. Brye

ABSTRACT

Genotypic differences for canopy wilting have been reported for soybean [*Glycine max* (L.) Merr.], but no wilting data have been published, and mechanisms for differences remain unresolved. In field studies in 2002 and 2003, differences for wilting among 19 genotypes were consistent across years. Plant introductions (PI 416937 and PI 471938) were among the slowest wilting genotypes, and breeding lines (93705-34 and 93705-95) were among the fastest wilting. Row spacing (18 vs. 80 cm wide) did not affect wilting, indicating that lateral rooting did not contribute to genotypic differences. In a separate field study, volumetric soil water content at 15- and 50-cm depths was generally greater for slow-wilting PI 416937 than for fast-wilting 93705-95. Wilting for both genotypes responded similarly to soil water content (r^2 = 0.63–0.74). In a growth chamber study, transpiration declined similarly for fast- and slow-wilting genotypes in response to soil water deficit. Wilting response to soil water was the same for slow-wilting PI 416937 and fast-wilting genotypes 93705-34 and A5959. Slow-wilting 93705-36 began wilting at a lower soil water content than did PI416937, 93705-34, and A5959, indicating that more than one mechanism may be responsible for slow wilting.

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Abbreviations: DAP, days after planting; FTSW, fraction of transpirable soil water; NTR, normalized transpiration ratio; PI, plant introduction; VPD, vapor pressure deficit.

Water deficit is the most common abiotic stress that reduces soybean [*Glycine max* (L.) Merr.] yields (Purcell and Specht, 2004). Due to prohibitive costs or a lack of available water source, irrigation is often not a viable option for alleviating water deficits in soybean production. According to Carter et al. (2006), there is little variation among U.S. soybean cultivars in their response to drought, although others (Mederski and Jeffers, 1973; Specht et al., 1986) have reported evidence of genetic diversity among cultivars for yield under drought conditions. Whether or not drought tolerance exists in the adapted North American breeding pool, little conscious selection pressure has been applied by breeders to improve this trait. Soybean breeding has traditionally involved selection for yield primarily in high-yielding environments, following the long-held assumption by breeders that genetic variation for drought tolerance is limited in the U.S. gene pool. Thus, development of drought-tolerant soybean cultivars appears to require identification of drought-tolerant germplasm and intentional incorporation into high-yielding genetic backgrounds.

Canopy wilting is the first visible symptom of soil water deficit in soybean. Soybean genotypes differ in the time of onset and the severity of canopy wilting in response to drought (Sloane et al., 1990;

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Carter et al., 1999). Field evaluations of plant introductions (PIs) from the USDA germplasm collection have identified PI 416937 and PI 471938 as slow wilting relative to commercially available cultivars (Carter et al., 2006). Understanding the mechanisms that contribute to differential canopy wilting is important for determining if delayed wilting could increase yields under drought and for determining under what conditions yields might be affected. Genotypic differences for canopy wilting may be associated with rooting traits, such as rooting depth, lateral rooting, or root-length density, that influence the ability to extract water from the soil profile. The slow-wilting genotype PI 416937 reportedly has a highly prolific root system (Pantalone et al., 1996) and increased lateral root growth (Hudak and Patterson, 1996) relative to other genotypes. In field studies, PI 416937 had greater root mass, root surface area, and root volume in the upper 38 cm of the soil profile than did the cultivar Forrest (Hudak and Patterson, 1995) and was able to maintain greater leaf turgor than other genotypes during imposed water deficits (Carter and Rufty, 1993). Comparison of eight cultivars with extremes for leaf-water potential indicated that high root density throughout the soil profile was associated with low midday leaf-water deficits (Boyer et al., 1980). Other studies indicated that a root-length density of 1.5 cm cm^{-3} was sufficient for maximum soil water extraction (Gardner, 1960; Cowan, 1965) and that soybean root-length density in the upper soil profile is generally several-fold greater than 1.5 cm cm^{-3} (Coale and Grove, 1990).

A second possible mechanism for slow wilting is soil water conservation. In a field study, Sloane et al. (1990) reported a numerically lower transpiration rate for the slow-wilting genotype PI 416937 than for Forrest under well-watered conditions. Fletcher et al. (2007) compared slow-wilting PI 416937 and PI471938 with commercial genotypes A5959 and Graham for whole-plant transpiration in response to vapor pressure deficit (VPD). A5959, Graham, and PI 471938 had maximum transpiration rates of 61 to 75 mg H_2O m^{-2} of leaf area s^{-1}, and transpiration increased linearly over the range of VPD evaluated (1–3 kPa). PI 416937 reached maximum transpiration of 42 mg H_2O m^{-2} s^{-1} at a VPD of 2.1 kPa, above which there was no additional increase in transpiration. These results indicate that PI 416937 may conserve soil moisture by limiting transpiration during periods of high evaporative demand.

There are no published data quantifying genotypic differences for wilting. For the slow-wilting trait to be useful for improving drought tolerance, it is important to demonstrate the trait's repeatability, to understand the conditions that lead to genotypic differences for wilting, and to understand the relationship between soil water content and canopy wilting. Previous unpublished field screens designed to identify slow-wilting genotypes have been conducted on 75- to 100-cm-wide rows (T.E. Carter, Jr., personal communication, 2007; C.H. Sneller, personal communication, 2007). Reports of greater lateral rooting for PI 416937 relative to Forrest (Hudak and Patterson, 1996) indicate that better root exploration of soil between soybean rows could account for genotypic differences in wilting in widely spaced rows. If so, the slow-wilting advantage for PI 416937 could be negated under narrow row spacings.

We hypothesized that genotypic differences for wilting were repeatable and resulted either from differences in soil water extraction due to differential rooting between fast- and slow-wilting genotypes or from conservation of soil water by slow-wilting genotypes. To test these hypotheses, the objectives of this research were to (i) evaluate differential wilting among soybean genotypes representing a large genetic base during progressive water-deficit stress; (ii) determine the influence of row spacing on wilting as an indicator of differential lateral rooting as a mechanism for wilting avoidance; (iii) evaluate canopy wilting and soil water extraction from various depths in the soil profile for fast- and slow-wilting genotypes; and (iv) compare fast- and slow-wilting genotypes for transpiration and wilting response to soil water content in a root-restricted environment.

MATERIALS AND METHODS

Experiment I: Row Spacing × Genotype Interaction Study

Field experiments were conducted at the Rice Research Experiment Station at Stuttgart, AR (34°30′ N, 91°33′ W), in 2002 and 2003 to evaluate the influence of row spacing and soybean genotype on canopy wilting in response to soil water deficit. The soil was a Crowley silt loam (fine, montmorillonitic, thermic Typic Albaqualfs) with organic matter content of 1%. Soil fertility was adjusted in accordance with the University of Arkansas soil test recommendations. The site was chosen because we observed that soybean wilted uniformly over a large land area at this site during a previous growing season.

The experimental design was a split plot with two row spacings (18 and 80 cm) as main plots and with 19 (2002) and 20 (2003) soybean genotypes as subplots with six replications. Soybean genotypes evaluated were either standard cultivars or genotypes chosen specifically because of prior experimental observations for wilting characteristics (Table 1). Experiments were planted on a flat seedbed on 6 June 2002 and on 10 June 2003 at a seeding rate of 43 seed m^{-2} in plots 6 m long. Narrow-row plots were seeded with a 9-row drill with 18 cm between rows for a plot width of 1.6 m. Wide-row plots had four rows spaced 80 cm apart for plot width of 3.25 m. Experiments were flood irrigated on 9 July 2002 and 16 July 2003, and no additional irrigation was applied during either growing season.

Canopy wilting between 1200 and 1500 h was rated visually on a scale of 0 to 100, where a wilting score of 0 represented no wilting; 20 represented slight wilting evidenced by leaf wilting and rolling in the top of the canopy; 40 represented somewhat severe leaf rolling in the top of the canopy, moderate wilting of leaves throughout the canopy, and some loss of petiole turgidity; 60 represented severe wilting of leaves throughout the

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Table 1. Description of 20 soybean genotypes evaluated in Experiment I, the row spacing × genotype interaction field study, conducted at Stuttgart, AR, in 2002 and 2003. Putative characteristics are based on unpublished data except where noted.

Genotype	Maturity group	Type of germplasm	Description
93705-17 93705-34 93705-78 93705-95	V	Recombinant inbred line	Fast wilting F_5 progeny of KS4895 × Jackson
93705-36 93705-37 93705-50 93705-80	V	Recombinant inbred line	Slow wilting F_5 progeny of KS4895 × Jackson
Isoline 53 Isoline 54	V	Recombinant inbred line	F_5 progeny of KS4895 × Jackson differing in a single wilting allele (Satt G138)
Isoline 55 Isoline 56	V	Recombinant inbred line	F_5 progeny of KS4895 × Jackson differing in a single wilting allele (Satt G138)
Jackson	VII	Cultivar	Old U.S. cultivar; good N_2 fixation under drought (Sall and Sinclair, 1991)
KS4895	IV	Cultivar	High-yielding U.S. cultivar; drought-sensitive N_2 fixation (Purcell et al. 1997)
Hutcheson	V	Cultivar	Stable yield over environments (Hufstetler et al., 2007)
Young	VI	Cultivar	Stable yield over environments (Hufstetler et al., 2007)
A5959†	V	Cultivar	High-yielding U.S. cultivar; fast wilting
N96-6809	VII	Breeding line	High-yielding grandchild of PI 416937 (Hufstetler et al., 2007)
PI 416937	V	Plant introduction	Slow-wilting introduction from Japan; prolific rooting (Mian et al., 1996)
PI 471938	V	Plant introduction	Slow-wilting introduction from Nepal; high yielding under stress; high WUE‡, low epidermal conductance (g_e) (Hufstetler et al., 2007)

†A5959 was evaluated in 2003 only.

‡WUE, water use efficiency.

canopy, with advanced loss of petiole turgidity; 80 represented plants with petioles severely wilted and dead leaves throughout much of the canopy; and 100 represented plant death. Canopy wilting was rated five times over a 6-d period between 1 August and 6 August in 2002. No other canopy wilting data were collected in 2002. Canopy wilting was rated five times between 13 August and 10 September in 2003.

Experiment II: Field Evaluation of Canopy Wilting and Soil Water Content

Additional experiments were conducted at the Rice Research Experiments Station at Stuttgart, AR, in 2003 and 2004 to evaluate differences in canopy wilting and volumetric soil water content for two genotypes during a progressive water-deficit stress. The two genotypes were chosen to represent wilting extremes from the row-spacing study in 2002 and were classified as slow wilting (PI 416937) or fast wilting (93705-95).

The site and cultural conditions were similar to those described for the row-spacing field study. Experiments were planted at a seeding rate of 43 seed m^{-2} on a flat seedbed on 10 June 2003 and 9 June 2004 in four-row plots with 80-cm spacing between rows and a plot length of 6 m. When soybean genotypes were at the VC growth stage (Fehr and Caviness, 1977) on 24 June 2003 and 15 June 2004, a hole was excavated (60 cm deep by 60 cm long) and water content reflectometers (Model CS616; Campbell Scientific, Logan, UT) for measuring volumetric soil water content were inserted horizontally into the soil 15, 30, and 50 cm below the soil surface. Water content reflectometer probes were oriented in the same direction as the soybean row and were placed 1 m from the end of each plot directly beneath an undisturbed portion of the one of the center two rows. To prevent damaging the probes, metal guides similar in dimension to the probe rods were inserted at the appropriate depth before probe installation. After probes were installed, cavities were filled, and seed of the appropriate genotype was sown to reestablish the soybean stands that were disturbed during probe installation. The reestablished seedlings were thinned to the desired density after emergence.

Canopy wilting was rated as described above during a progressive water deficit beginning 13 Aug. 2003 and 22 July 2004 until rainfall alleviated the water deficit. Volumetric soil water contents were measured using the probes and a data logger (Model CR-10; Campbell Scientific) each day that wilt ratings were recorded.

Experiment III: Growth Chamber Study

A growth chamber experiment was conducted at Fayetteville, AR, in 2004 to evaluate plant wilting and transpiration in response to a progressive water deficit in pots in which the root system was confined to a known soil volume. The experimental design was completely random with a factorial of four soybean genotypes (two slow-wilting and two fast-wilting genotypes as determined from the row-spacing experiment) and two soil water treatments (well watered and water deficit) with four replications.

Pots were constructed of 10-cm-diam., 40-cm-long polyvinyl chloride pipe, sealed at the bottom and top with end caps. A fitting was inserted into the bottom of each pot to allow for drainage. A 2-cm-diam. hole was drilled in the top through which the plant shoots emerged and through which water was applied to the soil surface. Pot tare weights were recorded, and pots were filled with a peat/perlite potting mixture (Lb2; Sun-Gro Horticulture, Bellevue, WA). The potting mix was saturated with deionized water, and then 1 L of a N-free nutrient solution (de Silva et al., 1996) was added to each pot. Pots were allowed to drain overnight and pot-capacity weights recorded. The term *pot-capacity weight* is used to describe the combined weight of the pot, soil mix, and water minus the tare weight of the pot and, therefore, represents only the weight of the soil mix and water. Pots were inoculated with *Bradyrhizobium japonicum* (strain USDA 110), and a single plant was grown through the hole in the top of each pot. The growth chamber was maintained at a constant 28°C with a light cycle from 0600 to 2200 h. Photosynthetically active radiation at the top of the plants was approximately 600 μmol m^{-2} s^{-1}.

Plants were well watered until 33 d after planting (DAP) by watering to 70% of pot-capacity weight as needed to keep soil

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water contents from falling below 50% of the capacity weight. At 1630 h on 33 DAP, all pots were watered to 70% of pot capacity. Four plants of each genotype were kept well watered by watering to 70% of pot-capacity weight at 1630 h daily from 34 to 42 DAP. The four plants for each genotype designated for the water-deficit treatment were watered at 1630 h daily to a target weight 50 g below that for the previous day, resulting in a slowly progressive water deficit. Pots were weighed at 0830, 1230, and 1630 h daily from 34 to 42 DAP. Wilting was rated visually as described for the field experiments for all plants each time pots were weighed. The change in pot weight between subsequent weighing times was assumed to be water lost through transpiration. The average pot weight for two subsequent weighing times was used as the pot weight associated with corresponding transpiration data. Transpiration values were calculated for three intervals (1630 to 0830, 0830 to 1230, and 1230 to 1630 h) for each 24-h cycle from 34 to 42 DAP. Transpiration values for each time interval were double normalized within each genotype to give a normalized transpiration ratio (NTR) ranging from 0 to 1 (Ray and Sinclair, 1997; Hufstetler et al., 2007). The first normalization corrected for differences in transpiration among plants before initiation of the water-deficit treatment, and the second normalization corrected for fluctuations in transpiration for control plants among measurement periods during the progressive water deficit. The NTR was approximately 1.0 for well-watered plants throughout the experiment.

Plant-transpirable water for each pot was defined as the difference between pot-capacity weight and pot weight when transpiration for water-deficit plants was <10% of the well-watered controls for a given measurement period (Sinclair and Ludlow, 1986). According to this criterion, transpirable water was zero at 28% of pot-capacity weight for all genotypes. Pot weights were converted to a fraction of the transpirable water (FTSW) as described by King and Purcell (2001) for comparing the response of wilting and transpiration to soil water content among genotypes. Plant fresh weights were not recorded and were not considered when determining pot weights or for calculating FTSW values.

Statistical Analysis

Canopy wilting data from Experiment I were subjected to ANOVA using a general linear model (SAS, version 8; SAS Inst., Cary, NC) by year, with row spacing as main plot factor, genotype as subplot, and rating date as sub-subplot. All factors were considered fixed effects in these analyses. Subsequent analysis included ANOVA by each rating date and correlation analysis to determine the consistency of wilting among genotypes for different rating dates and years.

Canopy wilting for Experiment II was also evaluated by ANOVA for each year, with genotypes as main plots and measurement dates as subplots. Volumetric water content data at each of three soil depths were analyzed by measurement date to evaluate differences between genotypes.

Normalized transpiration and wilting data from the growth chamber experiment (Experiment III) were evaluated using segmented linear regression (SAS, PROC NLIN) as a function of FTSW. This analysis provides a 95% confidence interval for intercept values and points where slope changes occur.

RESULTS

Experiment I: Row Spacing × Genotype Interaction Study

In 2002, there was 83 mm rainfall between 13 July and 18 July followed by a 3-wk period without significant rainfall (<10 mm). Slight canopy wilting (ratings from 0 to 20) was evident after 2 wk without rainfall on 1 August (data not shown). Canopy wilting was rated five times in a 6-d period between 1 August and 6 August in 2002. In 2003, wilting was rated six times between 13 August and 10 September. When rating date was included in the ANOVA model, there was no main or interacting effect of row spacing, indicating that wilting was similar between plots planted in 18- and 80-cm row spacings. There was a significant genotype × rating-date interaction both years (data not shown).

To further evaluate the effect of row spacing on wilting, ANOVA was run separately for each year and rating-date combination, with row spacing treated as the main-plot factor and genotype as the subplot. The genotype × row spacing interaction was nonsignificant for all rating dates. There was a significant main effect of row spacing only for the final rating in 2003 (Table 2), when wilt ratings averaged 44 and 41 for the 18- and 80-cm row spacings, respectively. These results confirm that, in general, wilting was not influenced by row spacing for any of the genotypes evaluated.

Table 2. Descriptive statistics for soybean canopy wilting averaged over row spacing and by row spacing in Experiment I (row spacing × genotype interaction study) at Stuttgart, AR, in 2002 and 2003.

Year	Days after planting	Rating date	Canopy wilting score					
			Averaged over row spacing				Row spacing†	
			Mean ± SE	Min.	Max.	Max. – Min.	Narrow	Wide
2002	56	1	13 ± 4	7	20	13	12	13
	57	2	16 ± 5	8	24	16	15	16
	59	3	26 ± 7	13	37	24	25	26
	60	4	30 ± 7	14	43	29	29	30
	61	5	35 ± 6	23	46	24	35	34
2003	64	1	7 ± 3	1	14	13	7	6
	69	2	33 ± 3	26	38	12	33	33
	72	3	31 ± 3	24	37	13	31	31
	76	4	32 ± 5	22	40	18	32	31
	91	5	36 ± 5	30	46	16	37	36
	92	6	43 ± 6	33	54	21	44	41*

*Significant difference for wilting between row spacings ($P = 0.05$) within a rating date.

†Canopy wilting averaged across genotypes within a rating date. Analysis of variance indicated a significant main effect of row spacing on canopy wilting only for Rating Date 6 in 2003.

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The main effect of genotype was significant ($P < 0.001$) for wilting at each rating date (data not shown). Since analysis of data by year and for each year and rating-date combination indicated no effect of row spacing on canopy wilting (with the exception of the final rating in 2003 as mentioned above), data were averaged over replications and row spacing for each year, rating date, and genotype combination. These data were analyzed using the SAS procedure PROC CORR to further evaluate the genotype × rating-date interaction that the initial split-split plot analysis indicated was significant. Descriptive statistics from this analysis are summarized in Table 2. Mean wilting score, averaged across genotype within a rating date, ranged from a low of 7 for Rating Date 1 at 64 DAP in 2003 to a high of 43 for Rating Date 6 at 92 DAP in 2003. The difference in wilting score (maximum – minimum) between slowest wilting and fastest wilting genotypes at a given rating date increased as the severity of wilting increased (Table 2). The Maximum–Minimum difference was 13 for the first rating date both years when the mean wilting scores were low (7–13). The Maximum–Minimum difference increased to 21 to 24 for the final rating date each year when the mean wilting scores ranged from 35 to 42.

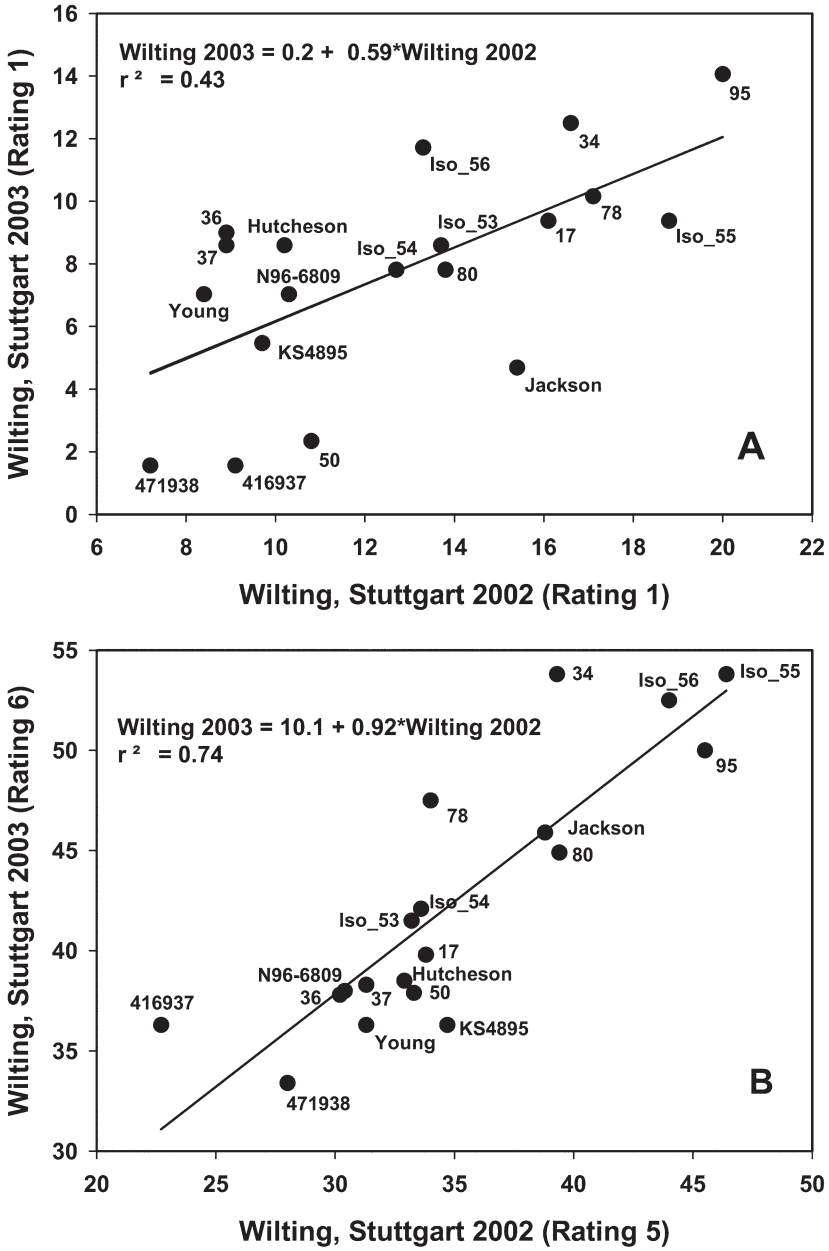


Figure 1. Visual ratings for soybean canopy wilting in Experiment I (row spacing × genotype interaction field study) at Stuttgart, AR, in 2002 and 2003 (A) early and (B) late in a progressive water deficit. Rating 1 was 56 and 64 d after planting (DAP) in 2002 and 2003, respectively. Rating 5 in 2002 and Rating 6 in 2003 were 61 and 92 DAP, respectively. Each data point represents the average of two row spacings (nonsignificant) and six replications for each year.

Rating Date 1 from 2003 had the lowest average canopy wilting (Table 2) and the weakest correlation with other rating dates ($r = 0.65$–0.79) (data not shown). Correlation coefficients for genotypic wilting between all other rating-date combinations ranged from $r = 0.72$ to 0.98 ($P < 0.0005$), indicating that genotypic differences for canopy wilting were consistent across years and rating dates. To demonstrate the consistency of relative wilting among genotypes, mean wilting score of each genotype for the rating dates with the lowest (Rating Date 1 from 2002 and 2003) and highest (Rating Dates 5 and 6 from 2002 and 2003, respectively) wilting scores for each year were plotted against each other (Fig. 1A and 1B). Regression indicated a relatively good linear relationship ($r^2 = 0.43$) for the mean genotypic wilting score between Rating 1 for 2002 and 2003 (Fig. 1A), when wilting was minimal each year. The linear relationship was stronger ($r^2 = 0.74$) between the final rating dates from each year when wilting was more severe (Fig. 1B). The relative ranking of wilting among genotypes remained fairly constant across years and rating dates (Fig. 1A and 1B). The two plant introduction lines, PI 416937 and PI 471938, were among the slowest wilting genotypes, and two recombinant inbred lines, 93705-34 and 93705-95, from the KS4895 × Jackson cross were among the fastest wilting lines at all rating dates. A5959 was also among the genotypes with highest wilting score at each rating date in 2003 (data not shown) but was not included in Fig. 1 because this genotype was evaluated only in 2003.

Experiment II: Field Evaluation of Canopy Wilting and Soil Water Content

Canopy wilting and volumetric soil water contents were recorded six times between 64 and

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92 DAP in 2003 and three times between 43 and 71 DAP in 2004 for the slow- and fast-wilting genotypes PI 416937 and 93705-95, respectively. Although there was a genotype × rating date interaction for wilting in both years, the fast-wilting line 93705-95 wilted more severely than PI 416937 at each rating date, and the magnitude of difference between genotypes was relatively consistent across dates (Fig. 2A and 2B). These results agreed with those from the row-spacing study in Experiment I, where, as canopy wilting increased in severity during the growing season, wilt ratings for 93705-95 were consistently 12 to 20 units greater than for PI 416937 (Fig. 1A and 1B).

PI 416937 had greater volumetric soil water content at the 15-cm soil depth for the first four dates in 2003 and the first two dates in 2004 (Fig. 2C and 2D). These differences were reflected in the lower wilting scores for PI 416937 at these rating dates. Later in the seasons, differences in soil water content were evident at the 50-cm depth but were not consistently in agreement with wilting response (Fig. 2A–2D). PI 416937 had greater soil water content at the 50-cm depth and wilted less than 93705-95 at 72 and 76 DAP in 2003 (Fig. 2A and 2C). At 71 DAP in 2004, PI 416937 wilted less than 93705-95 but had a lower soil water content at the 50-cm depth (Fig. 2B and 2D). Differences in volumetric water content in the upper soil profile were consistent with genotypic differences for wilting early in a progressive water deficit, but wilting difference did not always reflect differences in soil water content later in the season.

When wilting data were pooled over years and genotypes and regressed against volumetric water content at each of the three soil depths, there was a moderate to good (r^2 = 0.63–0.79) linear relationship (Fig. 3A–3C). These data indicate that genotypes PI 416937 and 93705-95 wilted similarly when there were similar amounts of soil water. Our original hypothesis was that slow-wilting genotypes either extract more water from the soil profile or conserve soil water relative to fast-wilting genotypes. These results indicate that the slow-wilting trait in PI 416937 is likely the result of a conservation of soil water.

Experiment III: Growth Chamber Study

The NTR and wilting data were plotted against FTSW for each genotype to evaluate the response of these traits to soil drying. Typical responses of NTR and wilting to changing FTSW are shown for the slow-wilting genotype 93705-36 (Fig. 4A and 4B). For 93705-36, NTR

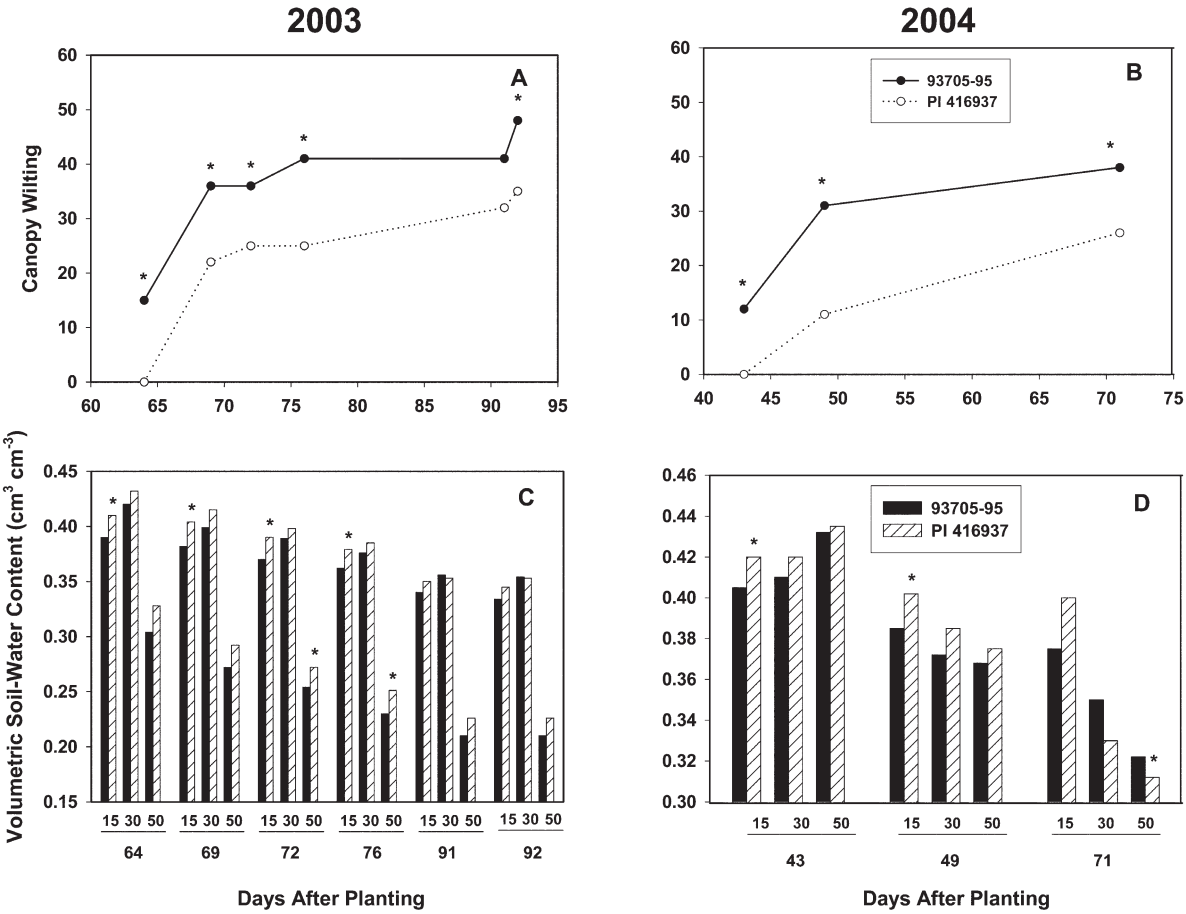


Figure 2. Soybean (A and B) canopy wilting and (C and D) volumetric soil water content for fast-wilting 93705-95 and slow-wilting PI 416937 in Experiment II conducted in the field at Stuttgart, AR, in 2003 and 2004. Asterisk (*) indicates a significant difference ($P < 0.05$) between genotypes within a rating date for wilting or between genotypes within a rating date and soil depth for soil water content. Soil depths (15, 30, 50 cm) are indicated below each pair of vertical bars in panels C and D.

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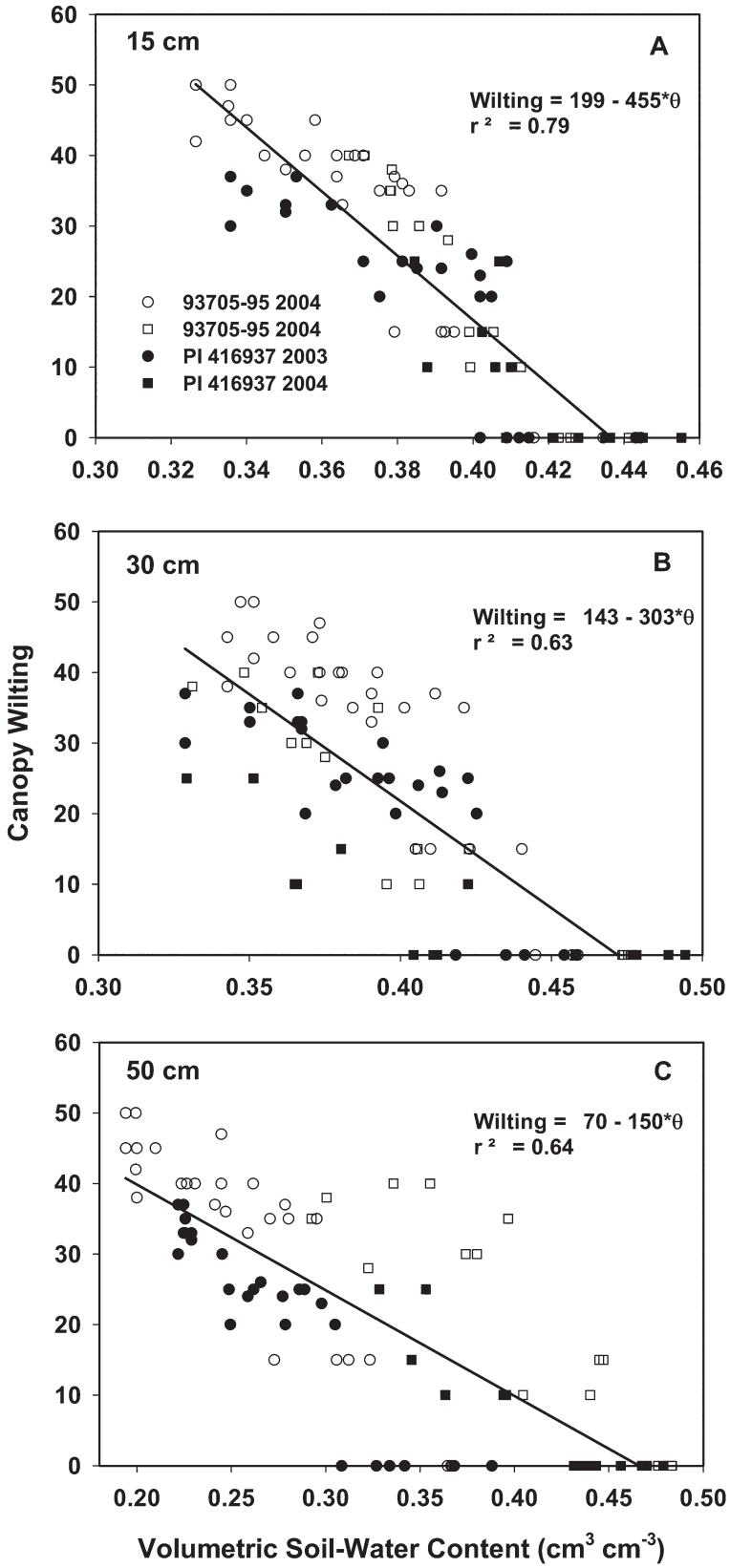


Figure 3. Wilting for slow-wilting (PI 416937) and fast-wilting (93705-95) soybean genotypes vs. soil water content (θ) at (A) 15-cm, (B) 30-cm, or (C) 50-cm soil depth in Experiment II at Stuttgart, AR, in 2003 and 2004. Each data point represents a single observation at a single rating date.

was equal to the well-watered control (value of 1) until FTSW reached a threshold value of 0.43 and declined linearly as soil water content declined below this FTSW value (Fig. 4A). Wilting was zero until FTSW reached 0.18 and then increased linearly as soil water content declined below this FTSW threshold (Fig. 4B).

In this experiment, plant fresh weights were not recorded and were not subtracted from pot weights for determining FTSW. This obviously introduced some error into the FTSW calculations. In a previous experiment with similar growing conditions and plant age, average plant fresh weights ranged from 39 to 44 g among five genotypes (data not shown). In the current experiment, the amount of transpirable water between pot-capacity weight and the 28% pot-capacity weight at the end of the experiment averaged 650 g. A 39- to 44-g plant fresh weight would account for approximately 5% of this transpirable water and would influence the FTSW values for transpiration and wilting break points. There were no visible differences in plant sizes among genotypes at the end of this experiment, and any minor differences in plant size would likely have little influence on differences in FTSW break points for transpiration and wilting among genotypes. For this reason, the current interpretation of these results should be valid for making comparisons among genotypes.

The FTSW thresholds where NTR began to decline in response to soil drying and the slope of the decline in NTR as soil dried below the threshold level were similar among fast- and slow-wilting genotypes (Table 3). Previous reports vary for the FTSW value at which transpiration begins to decline (Ray and Sinclair, 1998; Sinclair et al., 2000; Hufstetler et al., 2007). Sinclair et al. (2000) reported transpiration thresholds similar to ours (FTSW of 0.38–0.48) for eight PIs grown in a peat-based potting mix in pots approximately 75% as large as the ones used in this study. Transpiration thresholds for the soybean cultivar Biloxi ranged from FTSW of 0.33 for plants growing in 16.2-L pots to 0.37 in plants grown in 4.1-L pots (Ray and Sinclair, 1998). Hufstetler et al. (2007) reported transpiration thresholds at FTSW values of 0.19 to 0.28 for 23 soybean genotypes growing in a mineral soil. Of these publications, only Hufstetler et al. (2007) reported accounting for plant fresh weights when calculating FTSW. This may explain why FTSW threshold values reported by Hufstetler et al. (2007) were lower than others reported in the literature. Also the use of a mineral soil by Hufstetler et al. (2007) may have contributed to the lower FTSW values for transpiration thresholds as compared to those for experiments using peat-based potting mix.

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The slow-wilting line 93705-36 began wilting at a FTSW of 0.18, while all other genotypes wilted at significantly higher FTSW values (Table 3). There was no difference in the FTSW threshold for wilting or for the slope of wilting score vs. FTSW among the other three genotypes. The slope of wilting score vs. FTSW was also different for 93705-36 compared with the other genotypes. The more negative slope for wilting vs. FTSW for 93705-36 indicates that, although wilting began later in the soil drying cycle, wilting developed more quickly in response to soil drying below the wilting threshold FTSW than for other genotypes.

DISCUSSION

We hypothesized that slow-wilting soybean genotypes either extracted more soil water because of improved rooting traits, or they conserved soil water relative to fast-wilting genotypes before the onset of severe drought. Previously published data indicated prolific lateral rooting for the slow-wilting line PI 416937 (Pantalone et al., 1996). Row spacing did not influence canopy wilting in the present research, indicating that greater lateral rooting for slow-wilting genotypes was likely not responsible for genotypic differences in wilting. Wilting data from the field study in 2002 demonstrated that, for the Stuttgart field location, canopy wilting can progress rapidly (Table 2), but the general ranking (low to high) in wilting scores among genotypes remained relatively constant during the progression from slight to severe wilting in both years of our study (Fig. 1A and 1B).

Various mechanisms may contribute to slow wilting. Although our results indicated that slow wilting for PI 416937 was associated with soil water conservation, Sloane et al. (1990) reported that PI 416937 and the fast-wilting genotype Forrest had similar soil water content at 23-, 38-, and 53-cm depths in the soil profile during a progressive water deficit. Sloane et al. (1990) reported that the PI had a lower leaf osmotic potential and maintained greater leaf turgor, transpiration, and net carbon exchange than Forrest under stress. They concluded that osmotic adjustment may have contributed to the slow-wilting trait of the PI during water deficits. Fletcher et

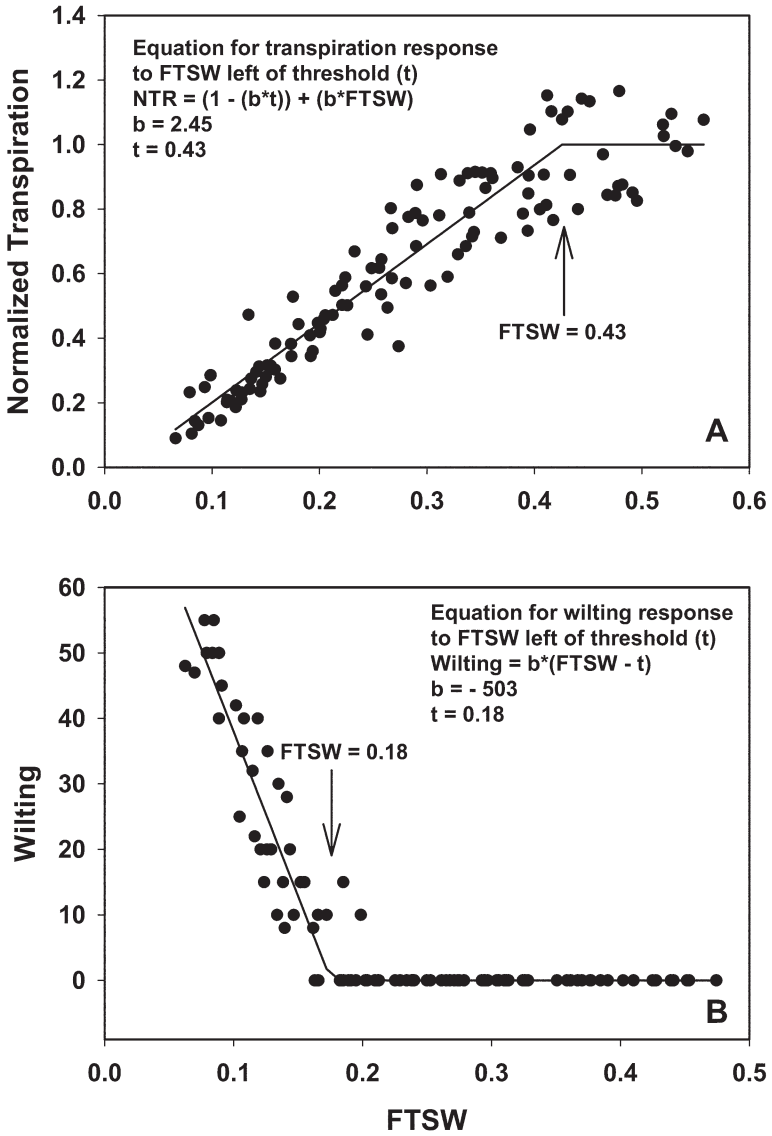


Figure 4. (A) Normalized transpiration ratio (NTR) and (B) wilting for the slow-wilting soybean line 93705-36 in response to the fraction of transpirable soil water (FTSW) in Experiment III (growth chamber study). Data were fit with the indicated segmented regression equations, each of which consist of a threshold FTSW value (*t*) above which there is a linear plateau where rates of the measured parameter are equal to the well-watered control and below which there is a linear response of the measured parameter to FTSW as described by the equations in panels A and B.

Table 3. Experiment III. Fraction of transpirable soil water (FTSW) thresholds below which wilting and transpiration differed for water-deficit plants relative to well-watered controls for soybean grown in a growth chamber. Slopes indicate the response of wilting and normalized transpiration ratios (NTR) to changing FTSW below the threshold values. Forms of the equations for wilting and NTR response to soil drying are given in Fig. 4.

Genotype	Wilting			Normalized transpiration	
	Type	Threshold FTSW	Slope	Threshold FTSW	Slope
PI 416937	Slow	0.26 ± 0.02†	–281 ± 30	0.41 ± 0.02	2.60 ± 0.20
93705-36	Slow	0.18 ± 0.01	–503 ± 50	0.43 ± 0.02	2.45 ± 0.20
A5959	Fast	0.25 ± 0.02	–266 ± 41	0.40 ± 0.02	2.43 ± 0.20
93705-34	Fast	0.23 ± 0.02	–308 ± 42	0.41 ± 0.03	2.68 ± 0.35

†± 95% confidence limits are given for each parameter estimate.

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al. (2007) reported that, under well-watered conditions, PI 416937 limited transpiration at VPDs above 2.1 kPa, whereas other genotypes continued to increase transpiration at higher VPD. Fletcher et al. (2007) speculated that such limitations on maximum transpiration rate could be a key trait for water conservation in less humid regions. Preliminary results evaluating biomass accumulation under well-watered conditions indicate that slow-wilting genotypes have lower radiation use efficiency than do fast-wilting lines (Purcell, 2006), supporting the hypothesis that slow wilting is associated with soil water conservation.

Results from our field and growth chamber studies indicated that wilting differences among genotypes may be due to conservation of soil water by slow-wilting genotypes. Fast- and slow-wilting genotypes wilted similarly in response to soil water content in the field (Fig. 3A–3C). When plants were grown in pots in a growth chamber, transpiration and wilting response to FTSW was similar for PI 416937 and two fast-wilting lines (Table 3), whereas the other slow-wilting line (93705-36) began wilting at a lower FTSW. This suggests that multiple mechanisms may be responsible for slow wilting. If so, there may be opportunities to pyramid genes for different mechanisms contributing to slow wilting.

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